Ehsan Suez

Infectious disease forecasting • Outbreak modeling • Public health analytics

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Professional Experience

2026–Present **Steven M. Teutsch Prevention Effectiveness Fellow**, *Centers for Disease Control and Prevention*
Center for Forecasting and Outbreak Analytics (CFA), Predict Division, Analytical Response Branch

- Conduct infectious disease modeling and outbreak analytics to support public health preparedness and response.
- Develop expertise in branching processes, renewal models, event-driven agent-based simulation, probabilistic programming, and modern Python/R workflows in Linux environments.
- Contribute to analytical work at the intersection of infectious disease dynamics, surveillance, forecasting, and public health decision-making.

2021–2026 **Graduate Research & Teaching Assistant**, *Bioinformatics, University of Georgia*.

2018–2020 **Graduate Research & Teaching Assistant**, *Biology, Missouri State University*.

Education

2026 **Ph.D. Bioinformatics**, *University of Georgia*, Athens, Georgia
Dissertation research: Optimizing forecasting pipelines for infectious disease dynamics
Advisor: Spencer J. Fox

2020 **M.S. Biology**, *Missouri State University*, Springfield, Missouri

2015 **B.S. Biotechnology & Genetic Engineering**, *Jahangirnagar University*, Bangladesh

Selected Research Projects

Forecasting **ScenarioCast: short-term forecasting from long-term epidemic scenarios.** Developed methods translating scenario projections into short-term influenza and COVID-19 forecasts.

Baselines **Optimized infectious disease baseline models.** Quantified how window length, transformations, drift assumptions, and other design choices affect forecast-hub benchmarks.

Nowcasting **Delay-adjusted influenza hospitalization nowcasting.** Developed machine-learning approaches to correct reporting delays and estimate recent burden in real time.

Variants **COVID-19 variant growth nowcasting.** Developed surveillance-based approaches for estimating growth of emerging SARS-CoV-2 variants.

Publications

Peer-Reviewed Publications

1. Mathis, S. M., Webber, A. E., León, T. M., et al.. Evaluation of FluSight influenza forecasting in the 2021–22 and 2022–23 seasons with a new target: laboratory-confirmed influenza hospitalizations. *Nature Communications*, 15, 6289. doi:10.1038/s41467-024-50601-9.
2. Islam, H., Khandker, S. S., Khatun, A., **Suez, E.**, Pranto, A. H., et al.. Hemodialysis tends to improve thyroid function by restoring hormone levels in ESRD patients compared to non-dialysis kidney disease patients: A case–control study. *Diseases*, 14(4), 128. doi:10.3390/diseases14040128.
3. Pranto, A. H., **Suez, E.**, Uddin, M. E., Tonmoy, H. S., Meem, M. M. R. M., Montasir, F., et al.. Global epidemiology of hepatitis B virus infection in dialysis patients: a systematic review and meta-analysis of prevalence, immunity, risk factors, and beyond. *International Urology and Nephrology*, 1–26.
4. Khandker, S. S., Jahan, S., Khan, A. A., Abrittee, B. F., Nanjiba, R., Islam, D. Z., **Suez, E.**, et al.. Global epidemiology of HIV among dialysis patients: A systematic review and meta-analysis. *Springer Nature* (In Press).

5. Shabbir, N. A., Kant, S. B., Rashid, K., et al.. Prevalence of HIV/AIDS among pregnant women in North America: A systematic review and meta-analysis. *Medicine*, 103, e40339.
6. Khandker, S. S., Jannat, N., Sarkar, D., Pranto, A. H., Intisar, A. H., Zaman, J., Uddin, M. N., & **Suez, E.**. Association between glomerular filtration rate and β-thalassemia major: A systematic review and meta-analysis. *Thalassemia Reports*, 13, 195–205.

Manuscripts Under Review

7. Davis, J. T., Kaur, G., Hines, A., ..., **Suez, E.**, et al. (2026). Multi-season evaluation and analysis of categorical trend forecasts of influenza hospital admissions in the United States. *Nature Communications* (Under Review). medRxiv preprint: 2026.08.31.26361843.
8. MacArthur, I., Robacker, T., Case, B., ..., **Suez, E.**, et al. (2026). Collaborative estimation and evaluation of SARS-CoV-2 variant nowcasting in the United States. *Nature Communications* (Under Review). arXiv preprint: 2606.07129.
9. **Suez, E.**, Fox, S. J., et al. (2026). Basic baseline model design choices can substantially influence performance in collaborative forecast hubs. *PLOS Computational Biology* (In Review). Preprint.

Manuscripts In Preparation

10. **Suez, E.**, Fox, S. J., et al. Scenariocast: Harnessing scenario projections for short-term influenza forecasting. (In Preparation).
11. Cramer, A., **Suez, E.**, Fox, S. J., et al. Spatial risk assessment for a hypothetical H5N1 or H7N9 pandemic in the United States. (In Preparation).

Conference Presentations

Nov 2025	"Scenariocast: Harnessing scenario projections for short-term influenza forecasting." *Epidemics Conference (Epidemics10)*
Nov 2023	"From awareness to action: Enhancing clinician preparedness and public health policy development for managing monkeypox (Mpox) outbreaks." *APHA Annual Meeting*
Oct 2023	"Comparing the forecasting performance of baseline infectious disease forecast models." *MIDAS Annual Meeting*
Oct 2022	"Transcriptomic analyses of *Mycobacterium bovis* infection reveal potential biomarkers and therapeutic targets: A systematic review." *Steeve Giguère Science of Veterinary Medicine Symposium*
Jun 2022	"Investigating zoonotic tuberculosis infection patterns in Mexico and their links to bovine tuberculosis in the US." *Ecology & Evolution of Infectious Diseases Conference*
May 2020	"Effect of dose-dependent Vps1 on membrane fusion dynamics." *Einhellig Graduate Interdisciplinary Forum*
Nov 2018	"Tiny bubbles: Explaining membrane fusion dynamics." *CNAS Symposium, Missouri State University (3-Minute Thesis)*
Oct 2018	"Novel function of Vps1 on membrane fusion." *Midwest Yeast Annual Conference, Northwestern University*

Professional Service

Feb 2026–Present	**Member, Trainee Committee**, Models of Infectious Disease Agent Study (MIDAS)
Sep 2025–Present	**Member**, Ecological Society of America Disease Ecology Section Student Committee
Jun 2024–Dec 2025	**Student Representative**, MIDAS Coordination Center Steering Committee

Honors & Awards

2025 **Future Faculty Fellow**, University of Georgia

2024 **Halloran Scholarship**, Summer Institute in Statistics and Modeling in Infectious Diseases — $2,790

2023 **Outstanding Teaching Assistant**, University of Georgia

2023 **Summer Research Grant for Doctoral Students**, University of Georgia — $1,500

2022 **Summer School Scholarship**, University of Washington — $1,000

2020 **Outstanding Presentation Award**, Missouri State University

2019 **Honorable Mention (4th Place)**, CNAS 3-Minute Thesis Competition

2018 **Biology Summer Research Assistantship**, Missouri State University — $2,668

2012–2016 **Jahangirnagar University Scholarship**, Merit-based

Professional Development

2024 **Graduate Certificate in Public Health**, University of Georgia

GradTeach Certificate in Teaching Development, University of Georgia

2022 **Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)**, University of Washington

Technical Skills

Programming R, Python, Bash; basic Rust

Methods Time-series modeling; infectious disease forecasting and nowcasting; statistical modeling; machine learning; branching processes; renewal models; event-driven agent-based simulation; probabilistic programming

Systems Linux/Unix environments; high-performance computing clusters; Git/GitHub; VS Code; `uv`; reproducible analytical workflows